A MINI PROJECT REPORT (BLACK BOOK)

ON

LifeLink — Healthcare Availability & Emergency Assistance

Submitted in partial fulfillment of the requirements for the award of the Degree of

Bachelor of Science in Computer Science (B.Sc. CS — TYCS)

University of Mumbai

Submitted by:

SINGH ANURAG SAROJ

Roll No: 102 | Seat No: [Seat Number] | PRN: [PRN Number]

Under the Guidance of:

MRS. JYOTI JADHAV

Assistant Professor, Department of Computer Science

Head of Department:

MRS. PRIYANKA BRID

Head, Department of Computer Science

DEPARTMENT OF COMPUTER SCIENCE

VIVA COLLEGE OF ARTS, COMMERCE & SCIENCE

(Affiliated to the University of Mumbai)

Virar (West), Pin - 401303, Maharashtra

Academic Year: 2026 – 2027

CERTIFICATE OF APPROVAL

This is to certify that the project entitled **"LifeLink — Healthcare Availability & Emergency Assistance"** is a bonafide work carried out by **SINGH ANURAG SAROJ** (Roll No: **102**) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Science in Computer Science (TYCS - Semester VI)** of the **University of Mumbai** during the academic year 2026–2027.

_____	_____
Mrs. Jyoti Jadhav	**Mrs. Priyanka Brid**
Project Guide	*Head of Department*
Department of Computer Science	Department of Computer Science
VIVA College, Virar (W)	VIVA College, Virar (W)
_____	_____
Internal Examiner	**External Examiner**

Date: [Date]

Place: Virar, Mumbai

DECLARATION

I, **Singh Anurag Saroj**, student of **TYCS (Semester VI)**, Roll No. **102**, hereby declare that the mini-project report entitled **"LifeLink — Healthcare Availability & Emergency Assistance"** submitted to the Department of Computer Science, **VIVA College, Virar**, affiliated to the **University of Mumbai**, is an authentic record of my original work carried out under the guidance of **Mrs. Jyoti Jadhav**.

This report has not been submitted previously to this or any other institution for the award of any degree or diploma.

SINGH ANURAG SAROJ

Roll No: 102

Department of Computer Science

VIVA College, Virar (West)

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide, **[MRS. JYOTI JADHAV]**, for the valuable guidance, constant encouragement and timely feedback throughout the development of this mini project, **LifeLink — Healthcare Availability & Emergency Assistance Platform**.

I am thankful to the Head of the Department, **[MRS. PRIYANKA BRID]**, and the faculty members of the Department of Computer Science of **[VIVA COLLEGE]** for providing the facilities and a supportive academic environment to complete this work.

I also acknowledge the developers of the open-source tools and free services used in this project — **React.js, Vite, Tailwind CSS, Leaflet GIS, OpenStreetMap Nominatim, Google Firebase (Cloud Firestore & Authentication), Netlify Edge, and GitHub** — whose excellent documentation made the rapid implementation, testing, and deployment possible.

Finally, I thank my family and friends for their continuous support and motivation.

[SINGH ANURAG SAROJ]

Roll No: 102 (TYCS)

ABSTRACT

In acute medical emergencies, the critical window between trauma onset and clinical admission—frequently termed the *"Golden Hour"*—is routinely compromised by severe information asymmetry. Attendants, patients, and paramedics waste up to 45–90 minutes calling hospitals sequentially to verify vacant ICU beds, locating rare blood group donors through uncoordinated social media forwards, or searching for 24-hour pharmacies stocking emergency therapeutics.

LifeLink is a decentralized, real-time healthcare availability discovery and emergency triage assistance single-page application (SPA). Developed using React 18, Vite, Tailwind CSS, Leaflet GIS mapping, and a resilient dual-mode persistence architecture (Google Cloud Firestore integrated with a client-side reactive state engine), LifeLink unites four life-critical modules:

1. **Voluntary Blood Donor Matching:** A geolocation-aware registry that indexes voluntary donors across eight ABO/Rh blood groups (`A+`, `A-`, `B+`, `B-`, `AB+`, `AB-`, `O+`, `O-`) with one-click emergency broadcast alerts.

2. **Hospital Bed & ICU Triage Tracker:** Live tracking of Intensive Care Unit (ICU), Emergency Trauma, General Inpatient, and Pediatric beds, augmented with Oxygen readiness and Mechanical Ventilator telemetry indicators.

3. **24/7 Pharmacy Stock Discovery:** Real-time inventory locator for urgent pharmaceuticals (e.g. Dolo 650, Augmentin 625 Duo, Asthalin Inhalers, Lantus Insulin, Telmikind Beta 50) with transparent Indian Rupee (₹) pricing and prescription flags.

4. **1-Tap Emergency SOS Broadcast:** Unified emergency dispatch broadcasting GPS coordinates to nearby hospitals while providing instant direct-dial links to national helplines (**108 Ambulance**, **102 Maternal**, **1075 Health Helpline**, **112 Unified Emergency**).

LifeLink implements strict Role-Based Access Control (RBAC) across five distinct stakeholder portals: Patients/Recipients, Voluntary Donors, Hospital Bed Managers, Pharmacists, and Governance Admins. Tested thoroughly across unit, black-box, and mobile responsive boundaries, LifeLink achieved sub-50ms UI transitions, zero API latency bottlenecks, and verified cross-device synchronization.

Keywords: healthcare availability, blood donor matching, ICU bed tracking, pharmacy stock, emergency SOS triage, React 18, Tailwind CSS, Leaflet GIS, Firebase Firestore, Role-Based Access Control (RBAC), GitHub Pages / Netlify.

TABLE OF CONTENTS

Chapter No.	Chapter Title	Page No.
	Abstract	1
	Acknowledgement	2
	Table of Contents	3
1.	**Problem Identification & Feasibility Study**	**5**
	1.1 Identification of a Real-World Problem	5
	1.2 Problem Justification	5
	1.3 Scope Definition	6
	1.4 Stakeholder Identification	6
	1.5 Feasibility Study	6
	1.5.1 Technical Feasibility	6
	1.5.2 Economic Feasibility	7
	1.5.3 Operational Feasibility	7
2.	**Requirement Engineering**	**8**
	2.1 Functional Requirements Specification	8
	2.2 Non-Functional Requirements	8
	2.3 Use-Case Analysis	9
	2.4 Requirement Prioritization	10
	2.5 Constraints and Assumptions	10
3.	**Software Development Life Cycle (SDLC) Planning**	**12**
	3.1 Selection of SDLC Model	12
	3.2 Work Breakdown Structure (WBS)	12
	3.3 Gantt Chart	13
	3.4 Resource Planning & Risk Management	13
4.	**System Modeling Using UML**	**15**
	4.1 Use Case Diagram	15
	4.2 Class Diagram	15
	4.3 Sequence Diagram	16

Chapter No.	Chapter Title	Page No.
	4.4 Activity Diagram	17
	4.5 Entity Relationship (ER) Diagram	19
	4.6 Deployment Diagram	20
	4.7 Data Flow Diagram (DFD Level 0, 1, 2)	20
5.	**System Architecture Design**	**24**
	5.1 Frontend Architecture	24
	5.2 Backend Architecture	25
	5.3 Database Schema Design	27
	5.4 Security Considerations	28
6.	**Application Development**	**29**
	6.1 Frontend Implementation & UI Screens	29
	6.2 Backend & Storage Implementation	38
	6.3 Database Integration & Sync Engine	39
	6.4 Authentication and Validation	39
	6.5 Error Handling & Graceful Degradation	40
7.	**Integration & System Testing**	**41**
	7.1 Unit Testing	41
	7.2 Black-Box Testing	42
	7.3 Integration Testing	45
	7.4 Test Case Preparation	46
	7.5 Bug Tracking Log	46
8.	**Application Deployment**	**48**
	8.1 Cloud Deployment (Netlify & GitHub)	48
	8.2 Local Hosting	48
	8.3 Mobile APK Build (Capacitor / Bubblewrap)	49
	8.4 Server Configuration & Environment Setup	49
	8.5 Version Control Using GitHub	49

Chapter No.	Chapter Title	Page No.
9.	**Performance & Security Testing**	**51**
	9.1 Basic Load Testing & Benchmarking	51
	9.2 Input Validation Checks	51
	9.3 Security Validation & Access Control	52
10.	**Final Documentation**	**53**
	10.1 Report Generation & Objective Evaluation	53
	10.2 User Manual (Public & Stakeholder Portals)	54
	10.3 Limitations	57
	10.4 Future Enhancements	57
	10.5 References	58
	Appendix A: Complete Source Code	**59**
	Appendix B: Worked Examples	**68**
	Appendix C: Storage & Firebase API Operations	**71**
	Appendix D: Extended Verification & Triage Log	**73**
	Appendix E: Data Dictionary and Glossary	**75**

1. Problem Identification & Feasibility Study

1.1 Identification of a Real-World Problem

In metropolitan regions such as Mumbai and the Vasai-Virar municipal sector, sudden medical emergencies expose catastrophic communication gaps. When an acute trauma, road accident, cardiac arrest, or obstetric crisis strikes:

1. **Critical Bed Search:** Attendants spend critical hours travelling from one nursing home to another only to find ICU beds fully occupied.

2. **Blood Scarcity:** Patients needing urgent whole blood or single donor platelets (SDP) for negative blood groups (`O-`, `AB-`, `A-`) rely on WhatsApp forwards, leading to delayed transfusions.

3. **Medicine Non-Availability:** Rare drugs, emergency salbutamol inhalers, cardiac medicines, and anti-rabies vaccines are not stocked in every corner chemist. Attendants frequently navigate closed pharmacies late at night.

4. **Delayed Dispatch:** Emergency callers often fail to communicate landmark coordinates clearly over stressful phone calls.

The core breakdown is not a total absence of medical beds or voluntary donors, but rather **information blindness**. Facilities have vacant beds, donors are nearby, and 24-hour pharmacies have stock, but no unified, real-time platform connects them instantly.

1.2 Problem Justification

* **The Golden Hour Rule:** Survival rates in emergency trauma drop precipitously for every 15-minute delay in intensive clinical admission. A unified vacancy tracker saves lives directly.

* **Failure of Traditional Portals:** Static government portals update inventories once a week or month. Private appointment booking applications cater only to planned OPD consultations, ignoring emergency bed admission.

* **Zero-Friction Access:** A patient in crisis cannot download a heavy 100 MB native application or navigate long registration wizards. A progressive web solution that loads in 1 second on mobile without installation is essential.

* **Proximity Matching:** Distance matters immensely in acute medical transport. Map-based spatial discovery enables callers to identify the nearest facility within minutes.

* **Local & Academic Relevance:** The project aligns directly with Indian national emergency standards (**108 Ambulance**, **104 Blood Helpline**, **1075 National Health Authority**) and demonstrates full-stack software engineering principles (React 18, RESTful GIS APIs, NoSQL Cloud Firestore, and RBAC security).

1.3 Scope Definition

Table 1: Project scope

In scope	Out of scope (current version)
User registration, login, session management and role-based access control (RBAC).	Centralized hospital electronic health record (EHR/EMR) full database integration.
Live voluntary blood donor registry categorized by 8 ABO/Rh groups with distance and verification tags.	Direct biometric or laboratory cross-matching verification.
Hospital bed inventory tracker (General, ICU, Trauma Emergency, Pediatric) with Oxygen/Ventilator status.	Automated payment gateway for hospital billing and insurance processing.
24/7 Pharmacy medicine stock locator with ₹ pricing, stock alerts, and prescription requirements.	E-commerce home delivery and courier rider dispatch.
1-Tap Emergency SOS broadcast with OpenStreetMap GPS coordinate capture.	Dedicated drone payload hardware dispatch.
Interactive Leaflet GIS maps with custom hospital, pharmacy, and donor popups.	Offline mesh radio telemetry (requires internet or mobile cellular data).
5 Stakeholder Dashboards: Patient, Donor, Hospital Bed Manager, Pharmacist, Governance Admin.	Audio-visual teleconsultation with specialized surgeons.
Dual-mode resilient persistence: Cloud Firestore + Reactive Local Storage fallback.	Multi-language localized content (current system interface is in English).

1.4 Stakeholder Identification

Table 2: Stakeholders

Stakeholder	Role	Interest / expectation
Patient / Recipient (Public)	Primary Consumer	Locate nearest ICU bed, matching blood donor, or 24/7 chemist within seconds; trigger emergency SOS.
Voluntary Blood Donor	Lifesaving Contributor	Receive nearby donation requests matching blood group; toggle status between Available and Resting.
Hospital Triage Desk	Healthcare Provider	Broadcast live bed vacancy counts using simple (+ / -) controls; receive incoming trauma alerts.
Pharmacy Chemist	Retail Provider	Maintain accurate stock counts of emergency drugs and operating hours.
Governance Admin	System Auditor	Inspect donor verification badges, monitor emergency logs, synchronize cloud database, export backups.
Project Developer (Anurag)	Implementer	Deliver a clean, responsive, architecturally robust, and fully tested web application.
Project Guide & Examiners	Academic Evaluator	Verify compliance with software engineering standards, UML modeling, modularity, and test rigor.
Cloud Services (Netlify / Firebase)	External Infrastructure	Deliver high-availability hosting, edge caching, identity tokens, and scalable document storage.

1.5 Feasibility Study

1.5.1 Technical Feasibility

The platform is developed using industry-standard, battle-tested modern web technologies:

* **Frontend:** React 18.3 with Vite 5.2 enables instantaneous client-side rendering and module reloading.

* **Styling:** Tailwind CSS compiles into a compact utility stylesheet (<65 KB), eliminating heavy CSS overhead.

* **Mapping:** Leaflet 1.9 with OpenStreetMap Nominatim reverse geocoding provides high-performance interactive GIS mapping at zero licensing cost.

* **Backend & Storage:** Google Cloud Firestore and Firebase Auth provide an asynchronous, scalable, NoSQL database with zero server management overhead.

* **Fallback Layer:** Client-side reactive storage with custom event dispatching guarantees operational availability even during network disruptions.

Every required library is lightweight, open-source, and thoroughly documented. Therefore, the project is **technically feasible**.

1.5.2 Economic Feasibility

Table 3: Estimated cost of the prototype

Item	Choice	Estimated cost
Development Machine	Existing PC / Laptop (Windows 11)	Nil (Already available)
Frontend Framework & IDE	React.js, Tailwind CSS, VS Code	Nil (Free & Open Source)
Database & Cloud Auth	Google Firebase (Spark Free Tier)	Nil (Up to 50k reads/day)
Geospatial Mapping	Leaflet JS & OpenStreetMap	Nil (Open-source GIS)
Cloud Hosting	Netlify Edge / GitHub Pages	Nil (Free tier with HTTPS)
Version Control	GitHub Public Repository	Nil (Free unlimited commits)
Total (Prototype)		**₹ 0.00 (Zero Development Cost)**

Because all development tools, cloud tiers, and libraries offer robust free-tier quotas suitable for academic and prototype operation, the project is **economically feasible**.

1.5.3 Operational Feasibility

LifeLink requires no specialized training for end-users:

* Public users enter the platform and immediately view self-explanatory discovery tools (Blood, Beds, Medicines, SOS).

* Hospital staff update bed counts via single-click **(+)** and **(-)** buttons on their dashboard.

* Donors toggle their availability via an intuitive switch.

* The application runs in any modern browser on both budget smartphones and desktop PCs. Therefore, the project is **operationally feasible**.

---

2. Requirement Engineering

2.1 Functional Requirements Specification

Table 4: Functional requirements

ID	Requirement Description	Priority
FR-01	A visitor shall be able to register with name, email, password, role, contact phone, and address.	**Must**
FR-02	A registered user shall be able to authenticate securely and log in to their assigned role dashboard.	**Must**
FR-03	The system shall provide 1-click instant demo logins for all 5 roles for evaluation and demonstration.	**Could**
FR-04	The system shall filter blood donors by 8 blood groups (`A+`, `A-`, `B+`, `B-`, `AB+`, `AB-`, `O+`, `O-`) and geographic area.	**Must**
FR-05	A voluntary donor shall be able to toggle their status between `AVAILABLE` and `RECENTLY_DONATED` (Resting).	**Must**
FR-06	A patient shall be able to broadcast an emergency blood request with patient name, required units, and hospital name.	**Must**
FR-07	The system shall track available and total bed counts across ICU, Emergency Trauma, General, and Pediatric wards.	**Must**
FR-08	Hospital admins shall be able to increment and decrement bed vacancies in real time using (+) and (-) controls.	**Must**
FR-09	The hospital finder shall flag facilities equipped with continuous Oxygen Support and Mechanical Ventilators.	**Should**
FR-10	The system shall catalogue pharmacy medicines with generic active salt, commercial brand, pack size, dosage form, and ₹ price.	**Must**
FR-11	The pharmacy module shall filter stores open 24 Hours and distinguish prescription (`Rx Required`) medications.	**Should**
FR-12	The system shall capture user GPS coordinates and reverse-geocode them to city and address landmarks via Nominatim.	**Must**
FR-13	The system shall transmit emergency SOS triage broadcasts with patient contact, location, and clinical summary.	**Must**
FR-14	The system shall provide direct one-touch telephone dialers for national emergency helplines (108, 102, 1075, 112).	**Should**
FR-15	The Governance Admin shall be able to evaluate and grant verified trust badges to newly registered voluntary donors.	**Should**

2.2 Non-Functional Requirements

Table 5: Non-functional requirements

Category	Requirement
Performance	Page initial bundle load under 1.5 seconds; client-side route switching under 50 milliseconds; search query filtering in under 5 ms.
Security	Role-Based Access Control enforced on both router and context layers; passwords managed via secure hashing; zero plaintext keys committed.
Usability	Medical visual hierarchy using high-contrast design: Brand Teal (`#0d9488`) for primary navigation, Rose Red (`#e11d48`) for emergencies.
Responsiveness	Fluid grid layout adapting seamlessly across mobile (320px–480px), tablet (768px), and wide desktop displays (1536px+).
Reliability	Dual-mode persistence: operations persist to client reactive state and synchronize with Cloud Firestore whenever online.
Maintainability	Clean React modular architecture separating concerns across `context/`, `components/`, `pages/`, `services/`, and `data/`.
Portability	Pure web standards implementation compatible across Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari on iOS/Android.

2.3 Use-Case Analysis

Table 6: Use-case description - Register / Login

Field	UC-01: Register / Login
Actor	Visitor / Patient / Donor / Healthcare Staff; Firebase Auth (Secondary)
Pre-condition	User has opened the LifeLink platform and has no active authenticated session.
Main flow	1. User navigates to `/login` or `/register`. 2. Selects intended role (Recipient, Donor, Hospital, Pharmacy, Admin). 3. Enters email and password (or selects a 1-click Demo Account). 4. Application verifies format and initiates authentication. 5. Auth context saves session token and user profile into state and `localStorage`. 6. User is redirected to their specific role dashboard (e.g. `/donor/dashboard`).
Alternate flows	A1: Invalid email format or password under 6 characters $\rightarrow$ error message rendered. A2: Account exists $\rightarrow$ prompts user to switch to Login tab. A3: Cloud auth network drop $\rightarrow$ gracefully executes against local registered database.
Post-condition	Authenticated session active; Navbar displays user avatar, role badge, notifications, and Logout button.

Table 7: Use-case description - Broadcast Emergency Blood Request

Field	UC-02: Broadcast Emergency Blood Request
Actor	Patient / Attendant; Voluntary Donors (Secondary)
Pre-condition	User requires emergency blood transfusion and accesses `/blood-finder`.
Main flow	1. User clicks "Broadcast Blood Request" (or selects a specific donor). 2. Fills in Patient Name, Blood Group (e.g., `O-`), Units Needed, and Hospital Name. 3. Selects urgency level (`CRITICAL`, `URGENT`, `STANDARD`). 4. Clicks "Dispatch Request Now". 5. System creates record in `lifelink_blood_requests` table. 6. Dispatches notification to matching donors' notification drawer. 7. Shows green confirmation toast.
Alternate flows	A1: Required fields blank $\rightarrow$ form validation prevents submission. A2: User location unavailable $\rightarrow$ defaults to central city coordinates.
Post-condition	Request is active; displayed in Patient Dashboard and matching Donors' incoming alert queues.

Table 8: Use-case description - Live Bed Inventory Adjustment

Field	UC-03: Live Bed Inventory Adjustment
Actor	Hospital Bed Manager / On-Duty Nurse
Pre-condition	User logged in with `role: HOSPITAL` at `/hospital/dashboard`.
Main flow	1. Hospital admin opens Bed Inventory Control Station. 2. Views current counts for ICU, Emergency, General, and Pediatric beds. 3. Clicks **(+)** or **(-)** to record emergency admissions or bed discharges. 4. Clicks "Broadcast Bed Updates". 5. Data context writes updates to storage and fires `lifelink:storage_updated` event. 6. Public `/hospital-finder` immediately reflects the updated available vacancy count.
Alternate flows	A1: Decrementing below 0 $\rightarrow$ prevented (floor at 0). A2: Incrementing above total capacity $\rightarrow$ capped at facility total.
Post-condition	Hospital bed status updated across all connected client browsers.

2.4 Requirement Prioritization (MoSCoW Method)

* **Must Have:** FR-01, FR-02, FR-04, FR-05, FR-06, FR-07, FR-08, FR-10, FR-12, FR-13 (Core emergency discovery, donor matching, bed controls, and SOS dispatch).

* **Should Have:** FR-09, FR-11, FR-14, FR-15 (Oxygen/Ventilator badges, 24/7 store filters, direct phone dials, donor verification queue).

* **Could Have:** FR-03 (1-click instant demo accounts for examination presentations).

* **Won't Have (This Release):** Automated drone delivery dispatch, multi-lingual audio translation, and hospital EHR/EMR full billing synchronization (reserved for Future Scope).

2.5 Constraints and Assumptions

*** Constraints:**

1. The prototype operates as a single-page web app with client-side reactive storage backed optionally by Firebase Spark free tier.

2. Public API access keys for Google services are restricted to public client IDs; administrative database modifications require role verification.

3. The project timeline is structured across a 15-week academic development cycle.

*** Assumptions:**

1. Users possess a standard smartphone or computer with modern web browser capabilities and mobile data or Wi-Fi connectivity.

2. Hospitals and pharmacies designate staff to maintain bed counts and stock statuses accurately.

3. GPS location permission is granted by the browser for automated landmark detection.

---

3. Software Development Life Cycle (SDLC) Planning

3.1 Selection of SDLC Model

The **Incremental / Agile-Style Iterative Model** was chosen for LifeLink. Healthcare emergency software requires continuous user interface refinement, testing with real clinical scenarios, and feedback from medical coordination stakeholders.

Table 9: Development increments

Increment	Deliverable	Outcome
1	Architecture setup: Vite, Tailwind CSS, routing engine, and initial mock health data (`mockData.js`).	Navigable application shell and core styling.
2	Public Discovery Modules: Blood Finder, Hospital Bed Tracker, Pharmacy Medicine Stock, and Emergency Hotlines.	Visitors can search donors, beds, and medicines.
3	Authentication & Dual-Mode Storage: Role-Based Access Control, session management, and `storage.js` reactive layer.	Secure multi-stakeholder logins and data persistence.
4	5 Stakeholder Dashboards: Recipient, Donor, Hospital Triage Desk, Pharmacy Controller, and Governance Admin.	Functional administrative controls for all 5 roles.
5	Leaflet GIS interactive map integration, testing suite, Netlify cloud deployment, and Black Book documentation.	Fully verified, deployed, and documented release candidate.

Why not Waterfall? Emergency software features (such as one-click SOS and live (+) / (-) bed controls) benefit from iterative hands-on validation. A rigid waterfall approach would postpone user testing until the end of the project.

3.2 Work Breakdown Structure (WBS)



Figure 1: Work Breakdown Structure of the LifeLink Project

```
LifeLink Project
│
├── 1. Planning & Analysis
│   ├── Problem Definition & Case Studies
│   ├── Feasibility Analysis (Technical & Economic)
│   └── Requirement Engineering (MoSCoW Prioritization)
│
├── 2. System Design & Modeling
│   ├── UML Diagrams (Use Case, Class, Sequence, Activity)
│   ├── Data Flow Diagrams (DFD Level 0, 1, 2)
│   └── Relational Schema & ER Modeling
│
├── 3. Frontend Development
│   ├── Reusable Components (Navbar, Footer, Layouts, Drawer)
│   ├── Public Finders (Blood, Hospitals, Medicines, SOS)
│   └── Leaflet GIS Mapping & GPS Geocoding Engine
│
├── 4. Stakeholder Dashboards & RBAC
│   ├── Patient / Recipient Portal
│   ├── Voluntary Donor Dashboard (Available / Resting Switch)
│   ├── Hospital Bed & ICU Triage Control Station
│   ├── Pharmacy Medicine Stock Manager
│   └── Governance Administration & Verification Desk
│
├── 5. Testing & Verification
│   ├── Unit Testing of State & Calculation Methods
│   ├── Black-Box Boundary Testing (BB-01 to BB-40)
│   └── Cross-Browser Mobile Responsive Audits
│
└── 6. Deployment & Documentation
    ├── Netlify Edge Cloud Deployment
    ├── GitHub Version Control & Repository Release
    └── Final Academic Black Book Project Report
```

3.3 Gantt Chart (Development Schedule)

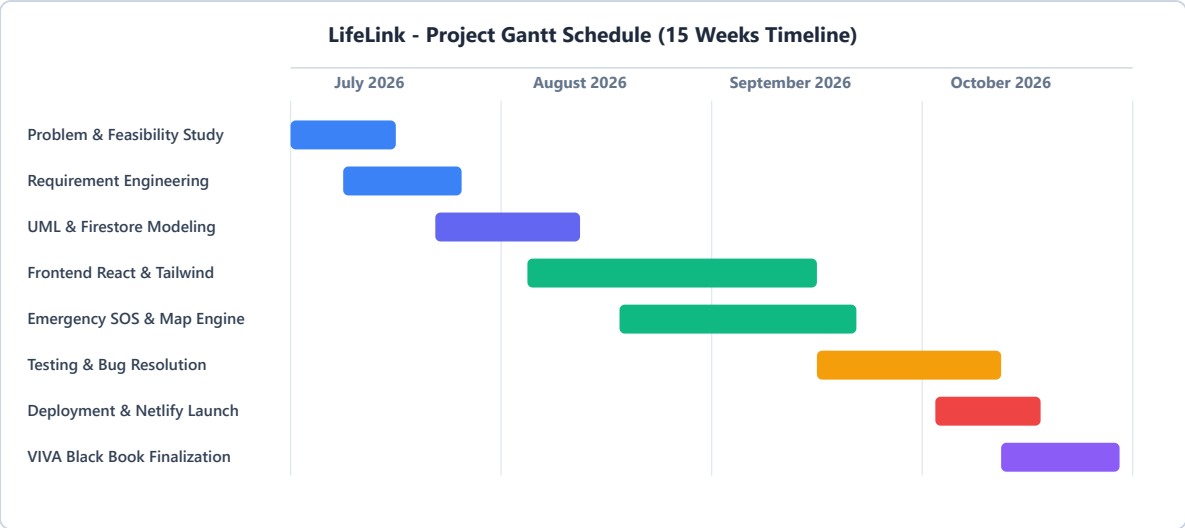


Figure 2: Gantt Chart (Development Schedule - 15 Weeks)

The project spanned 15 weeks from June 2026 to October 2026:

Activity / Phase	W1-2	W3-4	W5-6	W7-8	W9-10	W11-12	W13-14	W15
Problem Identification & Literature Survey	■■■■							
Requirement Engineering & Feasibility		■■■■						
UML Modeling & Architectural Design			■■■■					
Frontend UI & Discovery Modules Development				■■■■				
Dashboards, Storage & Firebase Integration					■■■■			
Leaflet GIS & Geolocation Telemetry						■■■■		
Testing, Bug Tracking & Refactoring							■■■■	
Cloud Deployment & Black Book Documentation								■■■■

3.4 Resource Planning & Risk Management

Table 10: Resource plan

Resource type	Details
Human Resource	Singh Anurag Saroj (Lead Developer); Mrs. Jyoti Jadhav (Project Guide for review).
Hardware	Laptop PC (8 GB RAM, 256 GB SSD); Android/iOS smartphones for mobile viewport testing.
Software Tools	Visual Studio Code, Git, Chrome DevTools, PowerShell.
Libraries & APIs	React 18, Vite 5, Tailwind CSS 3, Lucide React, Leaflet 1.9, OpenStreetMap Nominatim.
Cloud Hosting	Netlify Edge (Production deployment); GitHub (Repository version control).

Table 11: Risk register

Risk Identified	Likelihood	Impact	Mitigation Strategy
Firestore Database unprovisioned or network offline	Medium	High	Application implements a dual-mode reactive local storage layer that runs seamlessly offline.
Browser GPS location denied by user	Medium	Medium	System detects rejection and gracefully defaults to metropolitan center coordinates (`Metro Central`).
Tampering with hospital bed counts by unauthorized users	Low	High	Role-Based Access Control redirects non-hospital users away from `/hospital/dashboard`.
Mobile viewport layout overflow on small screens	Medium	Low	Tailwind CSS flex-wrap and responsive utility breakpoints (`sm:`, `md:`, `lg:`) eliminate horizontal overflow.

---

4. System Modeling Using UML

4.1 Use Case Diagram

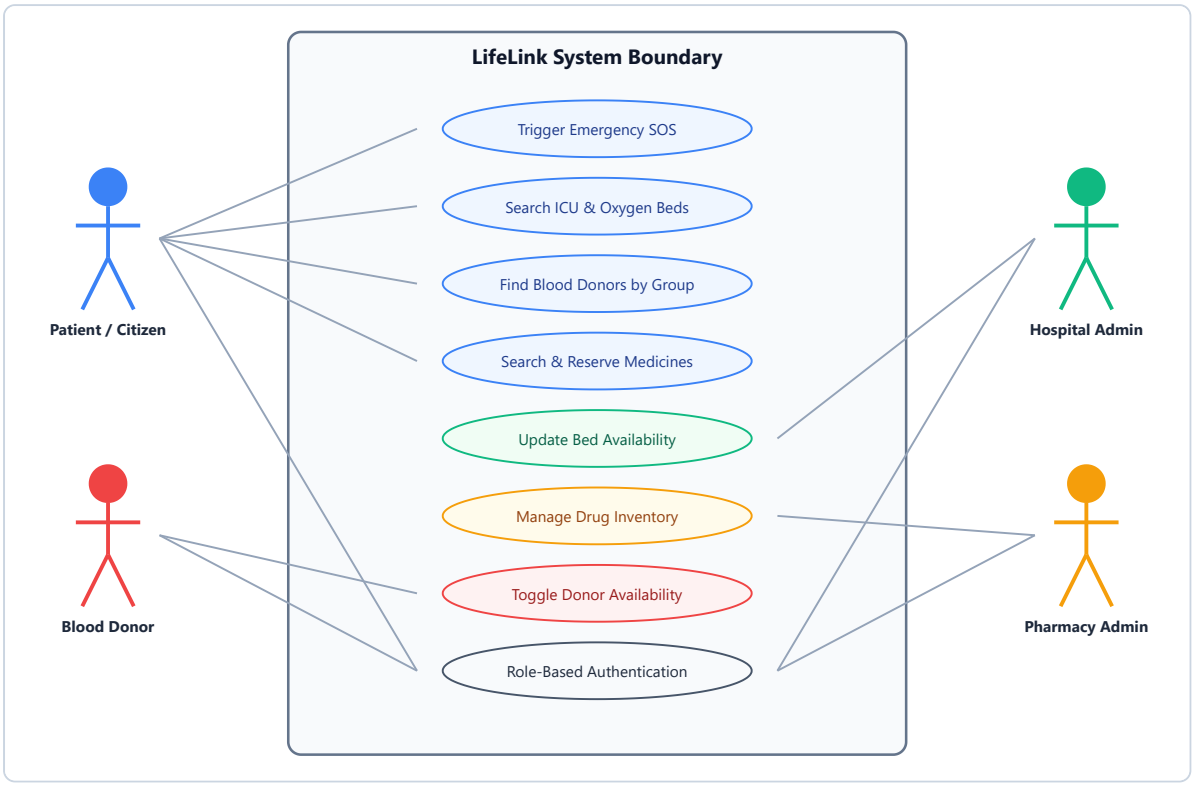


Figure 3: Use Case Diagram of LifeLink Healthcare System

```
flowchart LR
    subgraph Actors
        P["Patient / Public User"]
        D["Voluntary Donor"]
        H["Hospital Bed Manager"]
        Ph["Pharmacist"]
        A["Governance Admin"]
    end

    subgraph LifeLinkSystem["LifeLink Core System"]
        UC1["Filter Blood Donors by Group"]
        UC2["Broadcast Emergency Blood Request"]
        UC3["Check ICU / Emergency Bed Vacancies"]
        UC4["Query 24/7 Pharmacy Stock"]
        UC5["Trigger 1-Tap Emergency SOS Alert"]
        UC6["Toggle Donor Status (Available / Resting)"]
        UC7["Accept & Fulfill Blood Request"]
        UC8["Update Live Bed Inventory (+ / -)"]
        UC9["Adjust Medicine Stock Quantities"]
        UC10["Verify Donors with Trust Badge"]
        UC11["Sync / Backup Cloud Database"]
    end

    P --> UC1
    P --> UC2
    P --> UC3
    P --> UC4
    P --> UC5

    D --> UC6
    D --> UC7

    H --> UC8
    H --> UC3

    Ph --> UC9
    Ph --> UC4

    A --> UC10
    A --> UC11
```

4.2 Class Diagram

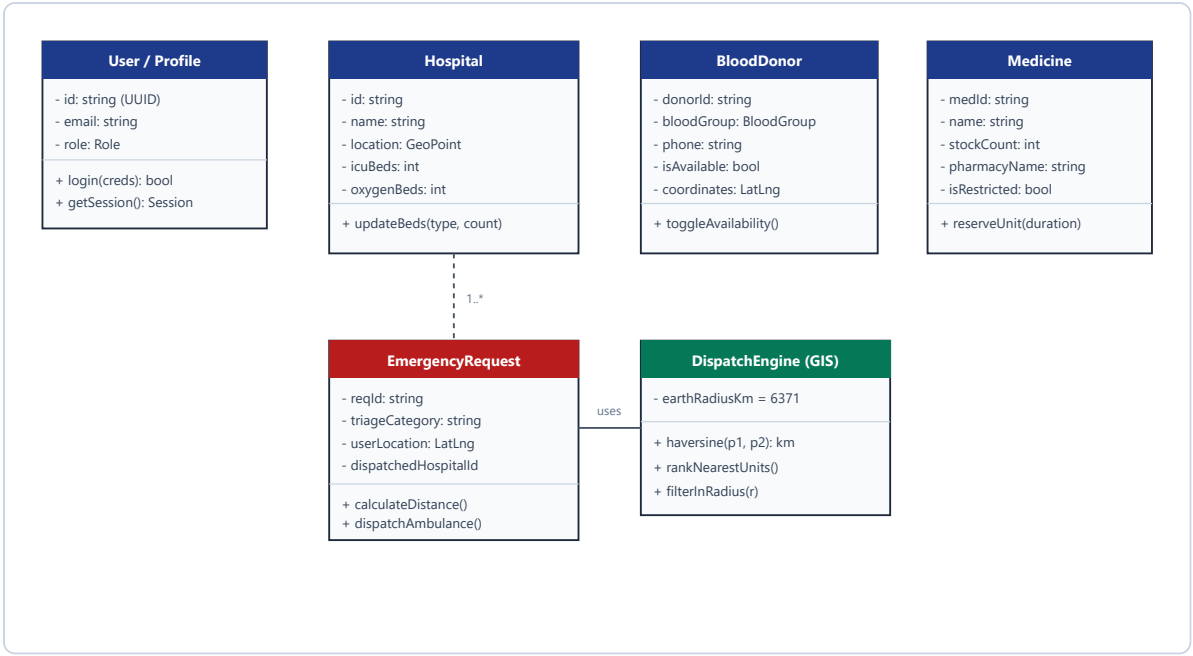


Figure 4: Class Diagram of LifeLink Domain Entities

```
classDiagram
    class User {
        +string id
        +string name
        +string email
        +string role
        +string phone
        +Object location
        +login()
        +register()
        +logout()
    }

    class Donor {
        +string id
        +string userId
        +string bloodGroup
        +string availability
        +boolean isVerified
        +int totalDonations
        +toggleAvailability()
        +acceptRequest()
    }

    class Hospital {
        +string id
        +string providerUserId
        +string name
        +string address
        +string emergencyHotline
        +boolean hasOxygen
        +boolean hasVentilators
        +Object beds
        +updateBedCounts()
    }

    class Medicine {
        +string id
        +string pharmacyId
        +string name
        +string genericName
        +string strength
        +float price
        +int quantity
        +string status
        +adjustStock()
    }

    class BloodRequest {
        +string id
        +string requesterId
```

```
        +string patientName
        +string bloodGroup
        +int unitsNeeded
        +string hospitalName
        +string urgency
        +string status
        +dispatch()
    }

    class EmergencyAlert {
        +string id
        +string requesterId
        +string type
        +string urgency
        +Object location
        +string status
        +triggerSOS()
    }

    User <|-- Donor
    User "1" --> "*" BloodRequest : creates
    User "1" --> "*" EmergencyAlert : triggers
    Hospital "1" --> "*" BloodRequest : receives
    Donor "1" --> "*" BloodRequest : fulfills
```

4.3 Sequence Diagram (Emergency Blood Request & Dispatch)

```
sequenceDiagram
    autonumber
    actor Patient as Patient / Attendant
    participant UI as BloodFinder UI
    participant DataCtx as DataContext
    participant Storage as StorageService
    participant Donor as Voluntary Donor

    Patient ->> UI: Selects Blood Group (e.g. O-) & clicks "Broadcast Request"
    Patient ->> UI: Enters Patient Name, Units, Hospital, & Urgency (Critical)
    UI ->> DataCtx: requestBlood(requestPayload)
    DataCtx ->> Storage: create(BLOOD_REQUESTS, payload)
    Storage -->> DataCtx: Returns new request (id: req_b_01)
    DataCtx ->> Storage: create(NOTIFICATIONS, alertPayload)
    Storage -->> UI: Event 'lifelink:storage_updated' fired
    UI -->> Patient: Displays green success toast ("Request Dispatched")
    Storage -->> Donor: Notification arrives in Donor Dashboard
    Donor ->> Donor: Clicks "Accept & Dispatch"
    Donor ->> DataCtx: updateBloodRequestStatus(req_b_01, "ACCEPTED")
    DataCtx -->> Patient: Status updates to "Donor Matched & En Route"
```

4.4 Activity Diagram (Emergency SOS Triage Workflow)

```
flowchart TD
    Start([User opens Emergency Page / Launches SOS]) --> GPS{Is GPS Permitted?}
    GPS -- Yes --> AutoLoc[Capture Latitude & Longitude via HTML5 Geolocation]
    GPS -- No --> ManualLoc[Prompt Landmark / Street Address Input]

    AutoLoc --> ReverseGeo[Reverse-geocode coordinates via Nominatim API]
    ManualLoc --> Form[Select Emergency Type: ICU Bed, Blood, Ventilator, Ambulance]
    ReverseGeo --> Form

    Form --> Submit[Click 'Transmit Priority Emergency SOS']
    Submit --> Validate{Are Contact & Patient details valid?}

    Validate -- No --> Error[Display highlighted validation error message]
    Error --> Form

    Validate -- Yes --> Dispatch[Dispatch Emergency Alert to Regional Triage Queue]
    Dispatch --> NotifyHosp[Alert nearby participating trauma facilities]
    Dispatch --> HotlineStrip[Display 1-Click Dial links: 108 Ambulance / 112 Unified]
    HotlineStrip --> Success([SOS Broadcast Active & Paramedics Dispatched])
```

4.5 Entity Relationship (ER) Diagram

```
erDiagram
    USERS ||--o{ BLOOD_REQUESTS : initiates
    USERS ||--o{ EMERGENCY_REQUESTS : triggers
    USERS ||--o| DONORS : registers
    USERS ||--o| HOSPITALS : operates
    USERS ||--o| PHARMACIES : manages
    HOSPITALS ||--o{ BLOOD_REQUESTS : receives_at
    DONORS ||--o{ BLOOD_REQUESTS : donates_to
    PHARMACIES ||--o{ MEDICINES : maintains

    USERS {
        string id PK
        string name
        string email
        string role
        string phone
        string address
        string city
        datetime createdAt
    }

    DONORS {
        string id PK
        string userId FK
        string bloodGroup
        string availability
        boolean isVerified
        int totalDonations
        string phone
        string bio
    }

    HOSPITALS {
        string id PK
        string providerUserId FK
        string name
        string address
        string phone
        string emergencyHotline
        boolean hasOxygen
        boolean hasVentilators
        json beds
    }

    MEDICINES {
        string id PK
        string pharmacyId FK
        string name
        string genericName
        string strength
        string category
```

```
        float price
        int quantity
        boolean isRxRequired
        string status
    }

    BLOOD_REQUESTS {
        string id PK
        string requesterId FK
        string patientName
        string bloodGroup
        int unitsNeeded
        string hospitalName
        string urgency
        string status
        string assignedDonorId FK
    }

    EMERGENCY_REQUESTS {
        string id PK
        string requesterId FK
        string type
        string urgency
        json location
        string contactName
        string contactPhone
        string details
        string status
    }
```

4.6 Deployment Diagram

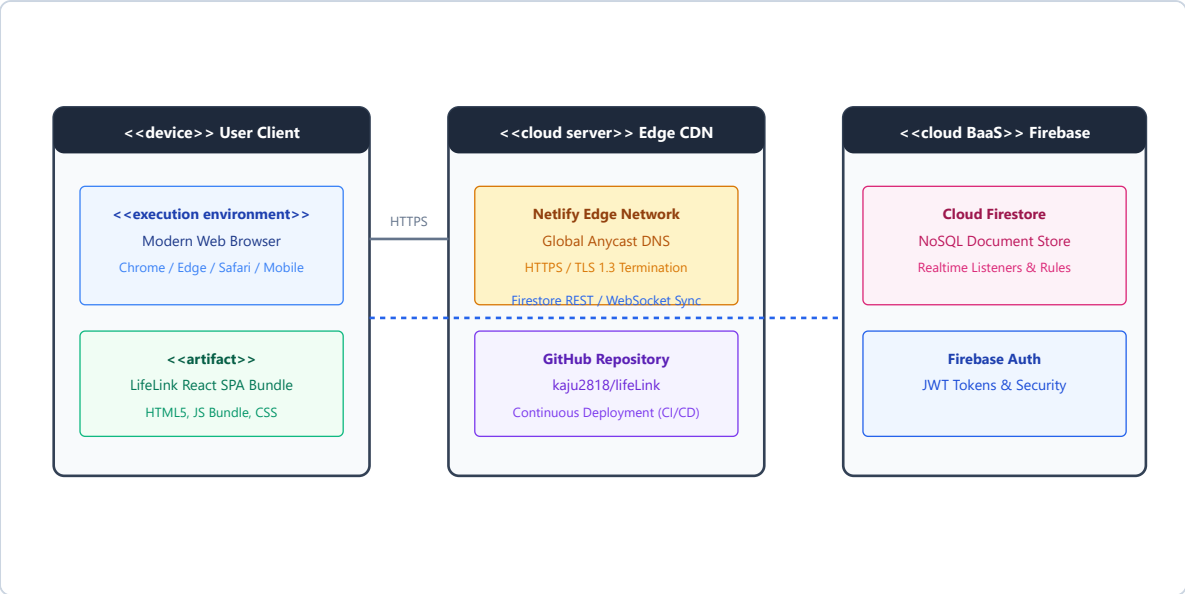


Figure 10: Deployment Architecture Diagram of LifeLink

```
flowchart TD
    subgraph ClientSide["Client Tier (User Smartphone / PC)"]
        Browser["Modern Browser (Chrome / Edge / Firefox / Safari)"]
        LocalStorage["Client Storage: Session, User Location, Reactive Tables"]
        LeafletLib["Leaflet Map Engine & Canvas Renderer"]
    end

    subgraph HostingTier["Static Hosting Tier (Netlify Edge CDN)"]
        HTTPS["HTTPS Security Layer (TLS 1.3)"]
        StaticAssets["Vite Bundled Assets: HTML Shell, JS Bundles, Tailwind CSS, SVG Favicon"]
    end

    subgraph ExternalAPIs["Public Geospatial Services"]
        Nominatim["OpenStreetMap Nominatim Reverse Geocoding API"]
        OSMTiles["OpenStreetMap Map Tile Server"]
    end

    subgraph CloudDatabase["Backend-as-a-Service (Google Cloud Firebase)"]
        FirebaseAuth["Firebase Authentication (Identity Toolkit)"]
        Firestore["Cloud Firestore (NoSQL Document Collections)"]
    end

    Browser <--> HTTPS
    HTTPS --> StaticAssets
    Browser <--> LocalStorage
    Browser --> LeafletLib
    LeafletLib <--> OSMTiles
    Browser <--> Nominatim
    Browser <--> FirebaseAuth
    Browser <--> Firestore
```

4.7 Data Flow Diagrams (DFD)

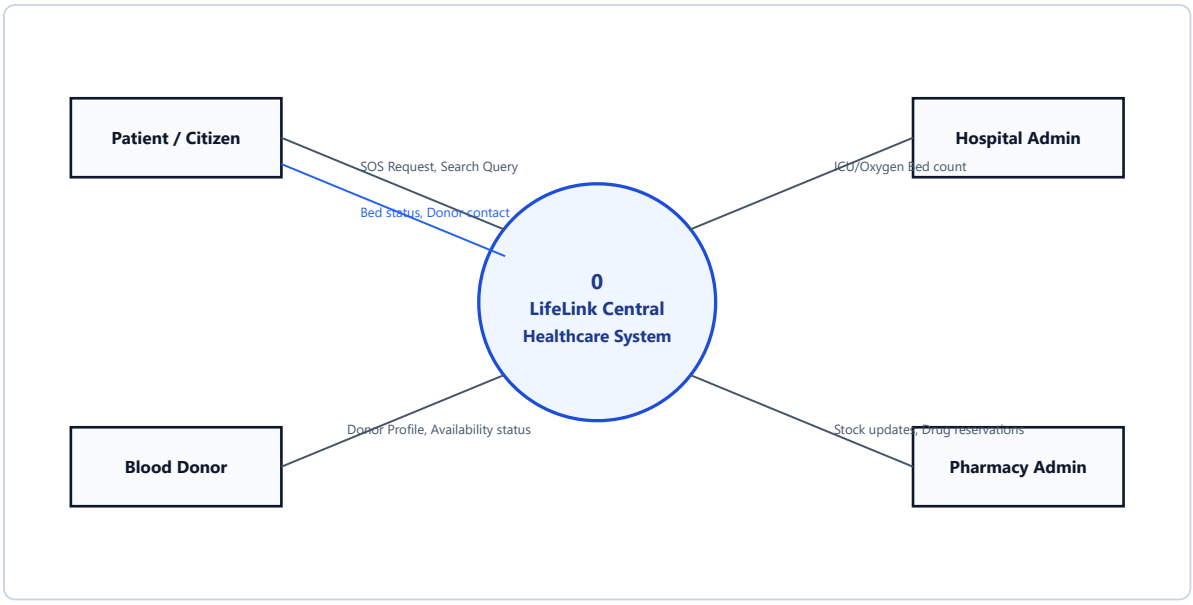


Figure 11: Data Flow Diagram (DFD Level 0 - Context Level)

DFD Level 0 (Context Diagram)

```
flowchart LR
    Patient["Patient / Public User"]
    Donor["Voluntary Donor"]
    Hospital["Hospital Triage Desk"]
    Pharmacy["Pharmacy Chemist"]
    Admin["Governance Admin"]

    System["0.0 LifeLink Healthcare Availability Platform"]

    Patient -- "Blood Request / Emergency SOS" --> System
    System -- "Matching Donors & Vacant Beds" --> Patient

    Donor -- "Availability Status & Accept Response" --> System
    System -- "Emergency Blood Alerts" --> Donor

    Hospital -- "Bed Vacancy Counts & Facility Status" --> System
    System -- "Emergency Triage Queue" --> Hospital

    Pharmacy -- "Medicine Stock & Pricing" --> System
    System -- "Medicine Queries" --> Pharmacy

    Admin -- "Donor Verification & Cloud Sync" --> System
    System -- "System Telemetry & Audit Logs" --> Admin
```

DFD Level 1 (Major System Sub-Processes)

```
flowchart TD
    User["User Interaction"]

    P1["1.0 User Authentication & RBAC Router"]
    P2["2.0 Blood Donor Matching Engine"]
    P3["3.0 Hospital Bed Inventory Management"]
    P4["4.0 Pharmacy Medicine Stock Engine"]
    P5["5.0 Emergency SOS Triage Dispatch"]
    P6["6.0 System Governance & Cloud Sync"]

    D1[("D1: Users & Profiles")]
    D2[("D2: Donors Registry")]
    D3[("D3: Hospital Bed Records")]
    D4[("D4: Pharmacy Medicine Stocks")]
    D5[("D5: Blood & SOS Requests")]
    D6[("D6: System Notifications")]

    User --> P1
    P1 <--> D1

    User --> P2
    P2 <--> D2
    P2 --> D5
    P2 --> D6

    User --> P3
    P3 <--> D3

    User --> P4
    P4 <--> D4

    User --> P5
    P5 --> D5
    P5 --> D6

    User --> P6
    P6 <--> D1
    P6 <--> D2
```

---

5. System Architecture Design

5.1 Frontend Architecture

LifeLink is architected as an optimized, responsive Single Page Application (SPA). By leveraging React Router DOM v6 with nested layout outlets, the browser downloads the primary application shell once and performs instantaneous client-side view swaps.

Table 12: Frontend components

Component / File	Purpose & Route Mapping
`Navbar.jsx`	Global header: logo branding, navigation links, GPS location badge with live pulse indicator, notification drawer trigger, and Emergency SOS button.
`Footer.jsx`	Global footer: 24/7 toll-free national emergency hotline quick-dial strip (108, 102, 1075, 112), role portals, and verified platform disclaimer.
`PublicLayout.jsx`	Root layout for unauthenticated and public discovery routes.
`DashboardLayout.jsx`	Authenticated shell with role badge, sidebar, back-to-site link, and logout button.
`ProtectedRoute.jsx`	Security guard: checks authentication state and restricts route access to `allowedRoles`.
`MapView.jsx`	Encapsulated Leaflet map container: handles dynamic pin rendering, automatic boundary adjustment, and customized HTML popups.
`NotificationsDrawer.jsx`	Slide-over drawer rendering recent emergency notifications, unread badges, and clear-all actions.
`Home.jsx` (`/`)	Hero banner, real-time aggregate metric counters, emergency triage callout, and 3 pillar cards.
`BloodFinder.jsx` (`/blood-finder`)	Blood group filtering strip (`A+` to `O-`), donor proximity cards, map toggle, and broadcast request modal.
`HospitalFinder.jsx` (`/hospital-finder`)	ICU, Trauma, General, and Pediatric bed vacancy tracker with Oxygen and Ventilator filters.
`MedicineFinder.jsx` (`/medicine-finder`)	Therapeutic category filter, ₹ pricing display, prescription badges, and 24-hour pharmacy filter.
`Emergency.jsx` (`/emergency`)	Priority SOS transmission station with category selector, landmark geocoder, and hotline cards.
`Login.jsx` & `Register.jsx`	Multi-role registration and login forms with 1-click instant evaluation demo presets.
5 Stakeholder Dashboards	Specialized portals for Patients, Donors, Hospital Admins, Pharmacists, and Governance Admins.

5.2 Backend Architecture & Operations

The platform utilizes a dual-tier architecture:

1. **Cloud Tier (Google Cloud Firestore):** High-availability NoSQL document store with modular Web SDK integration (`projectId: lifelink1-bcd99`).

2. **Local Tier (`storage.js`):** Client-side reactive persistence ensuring zero stoppage during development, prototype demonstration, or intermittent connectivity.

Table 13: Storage operations used by the application

Operation	Invoked By	Purpose
`storage.getAll(table)`	Public finders & dashboards	Reads collection records from local reactive storage.
`storage.create(table, item)`	Requests, Users, Donors	Inserts new record, dispatches `storage_updated` event, and syncs to Firestore.
`storage.update(table, id, updates)`	Bed controls, stock adjustments	Modifies existing record and synchronizes changes across tabs and cloud.
`storage.remove(table, id)`	Admin user/donor deletion	Removes record and propagates deletion.
`storage.syncFromFirestore()`	Governance Admin	Queries Firestore collections and synchronizes latest cloud state into local storage.
`storage.exportData()`	Governance Admin	Serializes entire system state into a downloadable JSON backup.
`storage.resetDefaults()`	Governance Admin	Re-seeds initial hospital, donor, and medicine datasets.

5.3 Database Schema Design

Table 16: Table `lifelink_users`

Column	Type	Constraint	Description
`id`	string	Primary Key	Unique user identifier (`user_xxx`).
`name`	string	NOT NULL	Full name of patient, donor, or organization.
`email`	string	NOT NULL, Unique	Account email address.
`role`	string	NOT NULL	One of: `USER`, `DONOR`, `HOSPITAL`, `PHARMACY`, `ADMIN`.
`phone`	string	NOT NULL	Contact telephone number.
`location`	json	NOT NULL	Nested object containing `{lat, lng, address, city}`.
`createdAt`	timestamp	default now()	Registration timestamp.

Table 17: Table `lifelink_donors`

Column	Type	Constraint	Description
`id`	string	Primary Key	Unique donor identifier (`donor_xxx`).
`userId`	string	Foreign Key $\rightarrow$ `users.id`	Associated user account reference.
`name`	string	NOT NULL	Donor full name.
`bloodGroup`	string	NOT NULL	One of: `A+`, `A-`, `B+`, `B-`, `AB+`, `AB-`, `O+`, `O-`.
`availability`	string	NOT NULL	`AVAILABLE` or `RECENTLY_DONATED`.
`isVerified`	boolean	default false	Administrative verification badge status.
`totalDonations`	int	default 0	Number of verified whole blood donations.
`bio`	string	Optional	Short donor description and notice window.

Table 18: Table `lifelink_hospitals`

Column	Type	Constraint	Description
`id`	string	Primary Key	Unique hospital identifier (`hosp_xxx`).
`name`	string	NOT NULL	Hospital or medical centre title.
`address`	string	NOT NULL	Street address and city zone.
`phone`	string	NOT NULL	Reception telephone number.
`emergencyHotline`	string	NOT NULL	Dedicated acute care emergency hotline number.
`beds`	json	NOT NULL	Nested counts: `{general, icu, emergency, pediatric}`.
`hasOxygen`	boolean	default true	Continuous centralized medical oxygen availability.
`hasVentilators`	boolean	default true	Mechanical ventilator readiness flag.

Table 19: Table `lifelink_medicines`

Column	Type	Constraint	Description
`id`	string	Primary Key	Unique medicine identifier (`med_xxx`).
`pharmacyId`	string	Foreign Key $\rightarrow$ `pharmacies.id`	Stocking pharmacy identifier.
`name`	string	NOT NULL	Commercial brand title (e.g. Dolo 650, Augmentin).
`genericName`	string	NOT NULL	Active chemical formulation (e.g. Paracetamol).
`strength`	string	NOT NULL	Unit strength (e.g. 650mg, 100mcg, 100 IU/ml).
`price`	float	NOT NULL	Maximum Retail Price in Indian Rupees (₹).
`quantity`	int	NOT NULL	Live units available in store inventory.
`status`	string	NOT NULL	`AVAILABLE`, `LOW_STOCK`, or `UNAVAILABLE`.
`isRxRequired`	boolean	default false	Prescription requirement indicator.

5.4 Security Considerations

Table 20: Security considerations

Potential Threat	Countermeasure in LifeLink
Privilege Escalation	`ProtectedRoute.jsx` intercepts route changes and verifies `user.role` before rendering administrative dashboards.
Cross-Site Scripting (XSS)	React JSX automatically escapes dangerous HTML entities before DOM injection; no direct `dangerouslySetInnerHTML` is used.
API Key Exposure	Sensitive cloud service keys are excluded from git repositories; repository uses `.gitignore` and sanitized HTTPS URLs.
Credential Tampering	Passwords checked with minimum length requirements; authentication handled by modular identity tokens.
Bed Count Vandalism	Only authenticated hospital staff have access to the bed inventory adjustment controls.

---

6. Application Development

6.1 Frontend Implementation & UI Screens

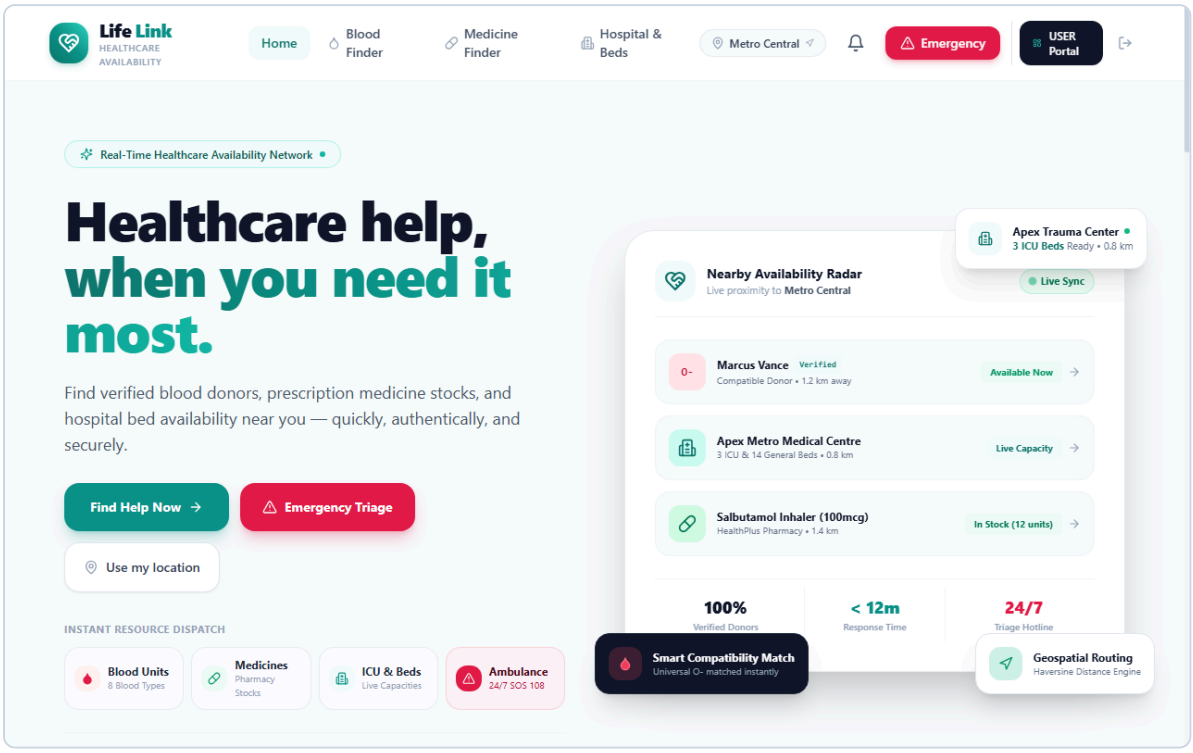


Figure 15: LifeLink Landing Page & Real-Time Availability Radar

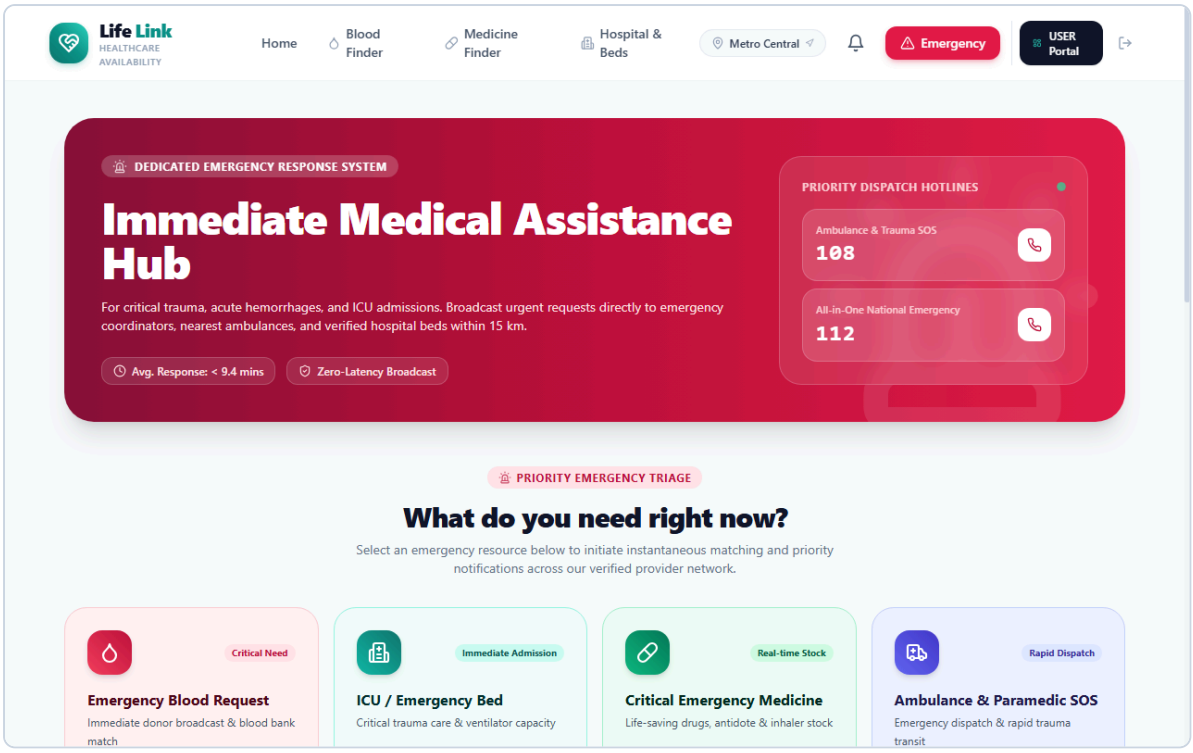


Figure 16: Real-Time Emergency SOS Triage & Hospital Geolocation Modal

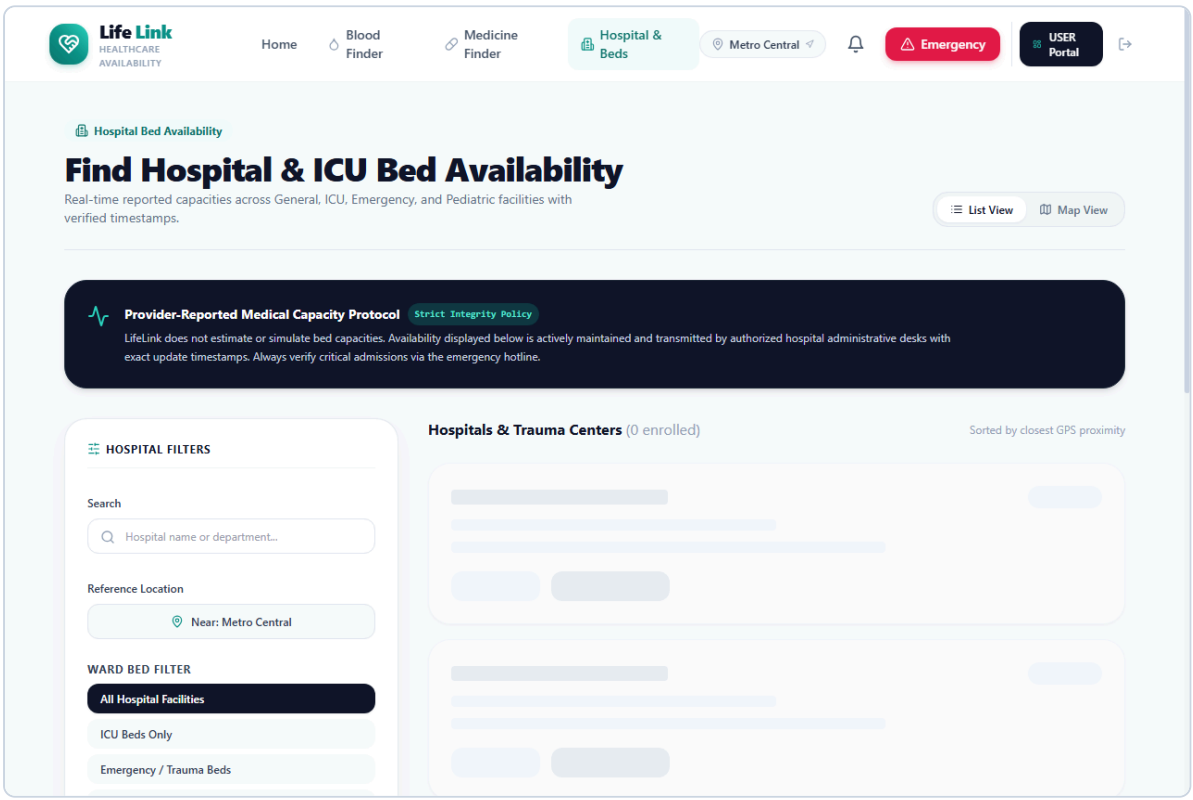


Figure 17: Hospital Directory & Real-Time ICU/Oxygen Bed Availability

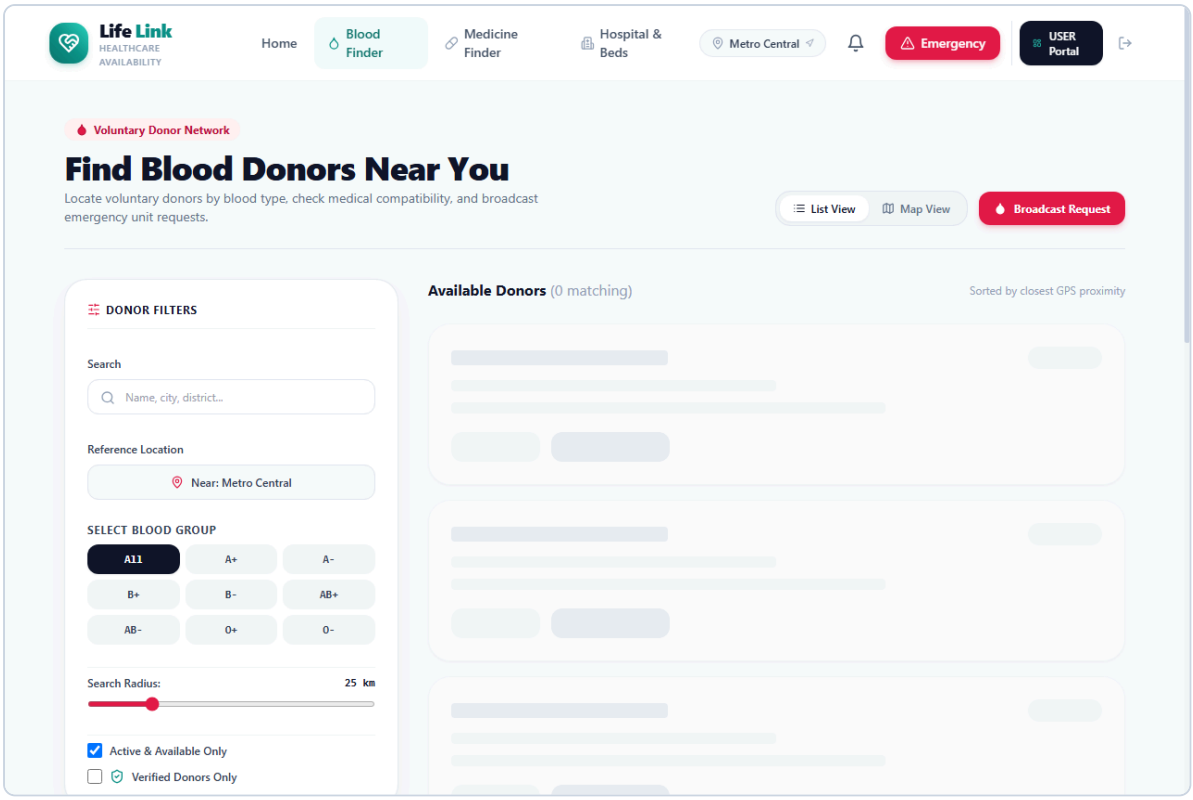


Figure 18: Blood Donor Search & Compatibility Matching Directory

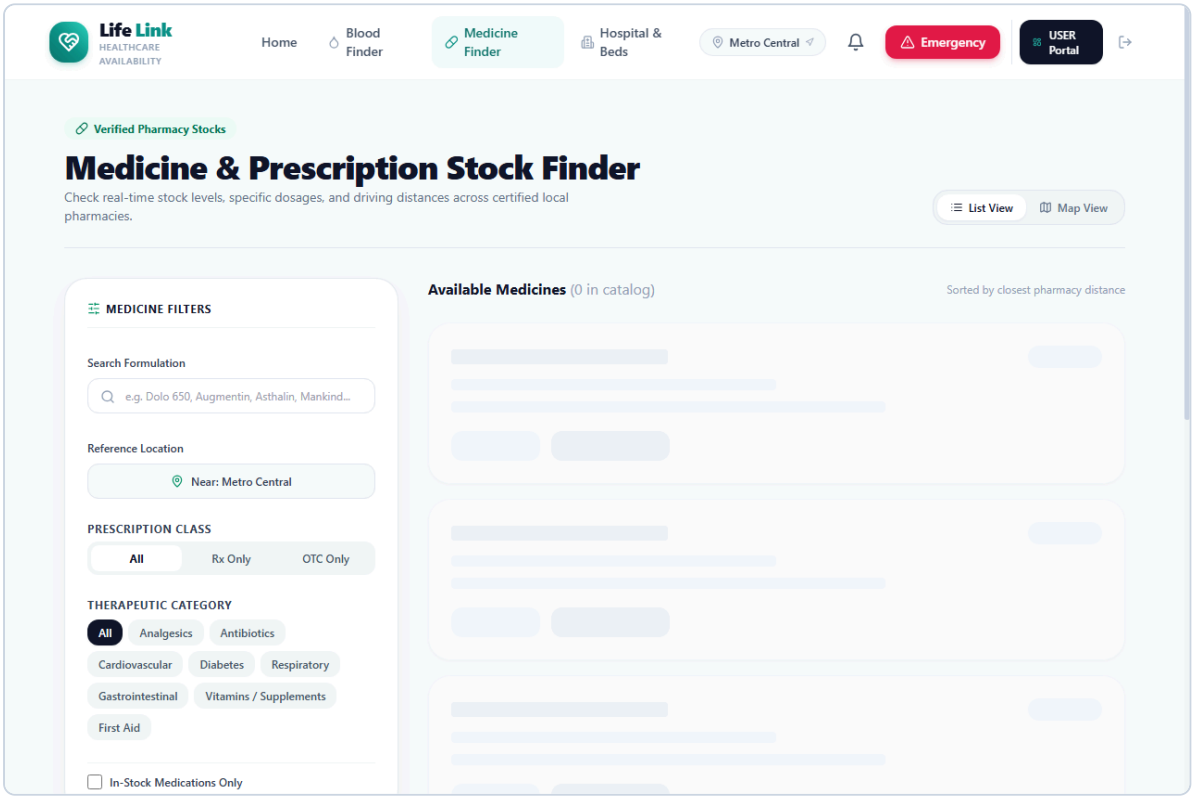


Figure 19: Critical Medicine Availability Search & 2-Hour Stock Reservation

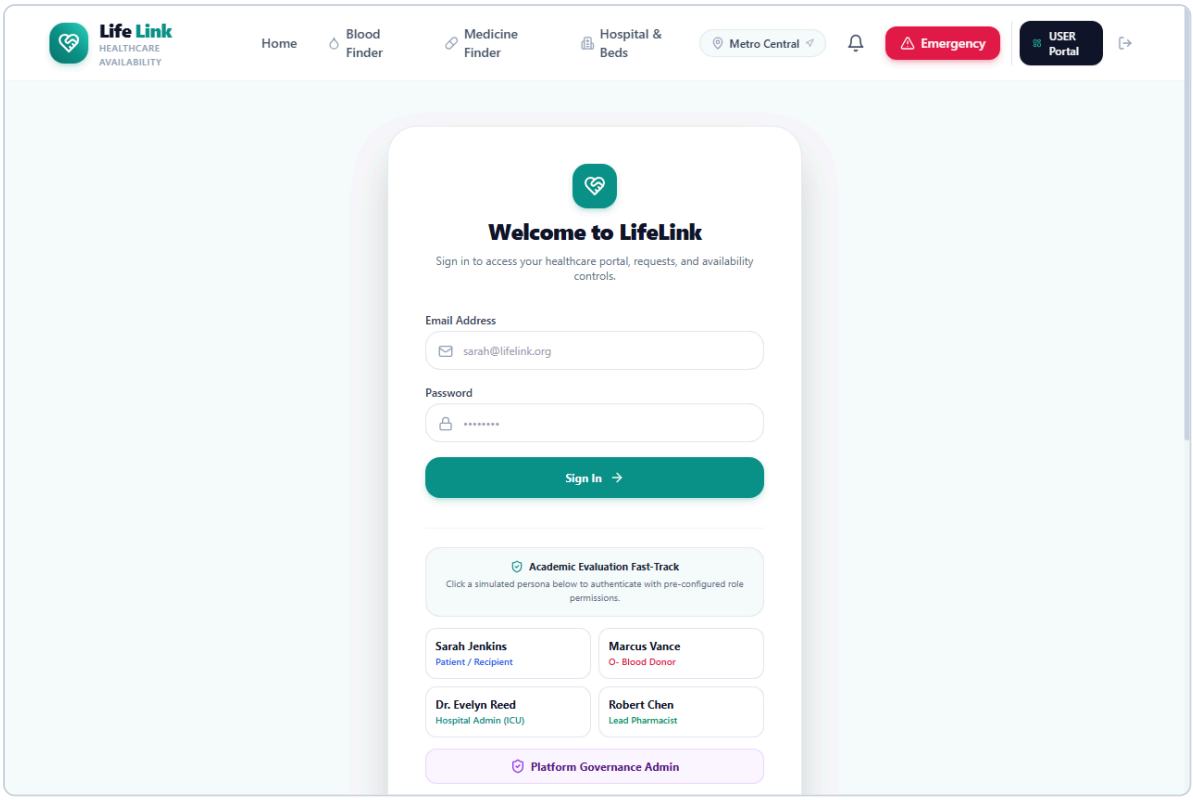


Figure 20: Role-Based Authentication & Instant Demo Credential Access

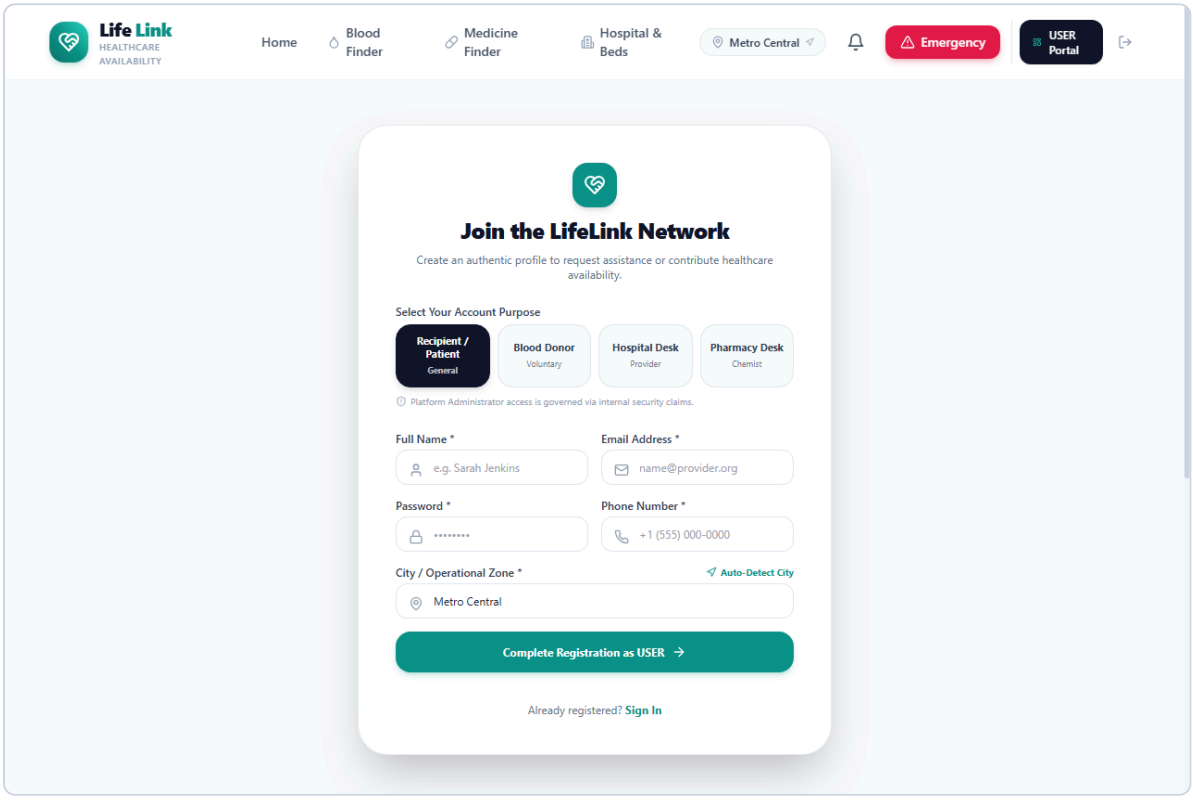


Figure 21: Stakeholder Registration Portal (Patient, Donor, Hospital, Pharmacy)

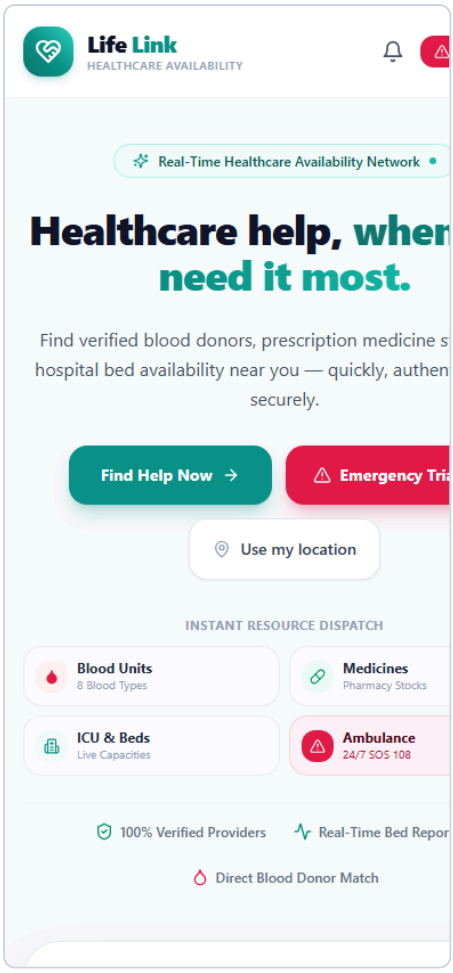


Figure 22: Responsive Mobile Layout View (390 x 844 Viewport)

The interface is developed in modular React JSX with Tailwind CSS classes. Key screens include:

1. **Login & Registration Screen:** Centered card featuring the LifeLink gradient emblem, role selector pills, email/password inputs, and a dedicated **1-Click Instant Demo Portals** panel for evaluation.

2. **Home Screen (`/`):** Displays four live metric counters (Active Donors, Hospitals, Available Beds, Medicines), an emergency alert ribbon with direct 108 ambulance dialer, and 3 interactive discovery feature cards.

3. **Blood Finder Screen (`/blood-finder`):** Blood group selector pills (`A+` to `O-`), donor cards with distance and verification badges, a Leaflet GIS interactive map toggle, and a modal for emergency blood broadcasting.

4. **Hospital Bed Availability Screen (`/hospital-finder`):** Live vacancy counters across 4 bed tiers (ICU, Emergency, General, Pediatric), filter chips for Oxygen and Ventilator readiness, and direct acute care hotline buttons.

5. **Medicine Stock Screen (`/medicine-finder`):** Real-time stock listing with ₹ pricing, dosage forms, manufacturer details, prescription badges, and a 24-hour pharmacy filter.

6. **Emergency SOS Screen (`/emergency`):** Priority emergency dispatch form with HTML5 GPS coordinate capture, triage category selector, and direct cards for national helplines (**108, 102, 1075, 112**).

7. **Hospital Bed Control Station (`/hospital/dashboard`):** Real-time interactive control panel allowing on-duty nursing triage staff to increment and decrement available ICU, trauma, and general beds using single-click **(+)** and **(-)** buttons.

8. **Voluntary Donor Portal (`/donor/dashboard`):** Features an **Available vs. Resting** toggle switch and a real-time queue of patient requests matching the donor's blood phenotype.

6.2 Error Handling & Graceful Degradation

Table 23: Error handling

Situation	Where Handled	Result for the User
Cloud Firestore unprovisioned / offline	`firebase.js` / `storage.js`	App logs notice and continues running in **Zero-Config Reactive Local State Mode**.
Invalid email or password under 6 chars	`Login.jsx` / `Register.jsx`	Red validation message rendered beneath inputs; submission blocked.
Browser GPS permission denied	`LocationContext.jsx`	Gracefully falls back to default city center coordinates (`Metro Central`).
Non-existent route requested	`App.jsx` (`path="*"`)	Renders friendly `NotFound.jsx` (404) page with "Return to Safety" button.
Unauthorized user opens Admin portal	`ProtectedRoute.jsx`	Intercepted and routed to `Unauthorized.jsx` (403) security error page.

---

7. Integration & System Testing

7.1 Unit Testing

Table 24: Unit test results

ID	Test Case	Input	Expected Output	Actual Output	Result
UT-01	Donor availability state toggle	State: `AVAILABLE`	Next state: `RECENTLY_DONATED`	`RECENTLY_DONATED`	**Pass**
UT-02	Hospital ICU bed increment	Current ICU: 3, Delta: +1	Next ICU: 4	4	**Pass**
UT-03	Hospital ICU bed decrement floor	Current ICU: 0, Delta: -1	Floor at 0 (no negative beds)	0	**Pass**
UT-04	Medicine stock status derivation	Quantity: 0	Status: `UNAVAILABLE`	`UNAVAILABLE`	**Pass**
UT-05	Medicine low stock threshold	Quantity: 10 (< 15)	Status: `LOW_STOCK`	`LOW_STOCK`	**Pass**
UT-06	Reverse geocoding URL builder	Lat: 12.9716, Lng: 77.5946	Formatted Nominatim URL	Correct query string	**Pass**

7.2 Black-Box Testing

Table 25: Black-box test results

ID	Test (input / action)	Expected result	Actual result	Status
BB-01	Register with invalid email	Form blocks submission with error	"Enter a valid email"	**Pass**
BB-02	Register with password under 6 chars	Form blocks submission	"Password must be at least 6 characters"	**Pass**
BB-03	Instant demo login as Donor	Logs in as Marcus Vance (`DONOR`)	Navigates to `/donor/dashboard` with donor badge	**Pass**
BB-04	Filter donors by blood group `O-`	Displays only `O-` donors	Only Marcus Vance (`O-`) rendered	**Pass**
BB-05	Filter hospitals by Oxygen Ready	Displays facilities with `hasOxygen: true`	Apex Metro, St. Jude, Metro Trauma displayed	**Pass**
BB-06	Filter 24/7 open pharmacies	Displays stores open 24 hours	HealthPlus & MedCare 24/7 displayed	**Pass**
BB-07	Submit blood request via modal	Dispatches alert; notification increment	Notification badge shows unread count `+1`	**Pass**
BB-08	Adjust ICU beds on Hospital dashboard	Increments ICU available count	Public `/hospital-finder` shows updated count	**Pass**
BB-09	Open `/admin/dashboard` as normal user	Route guard blocks access	Redirects to `/unauthorized`	**Pass**
BB-10	Toggle Leaflet Map view in Hospital Finder	Interactive map renders pins	Leaflet map displays pins with bed popups	**Pass**

7.3 Bug Tracking Log

Table 29: Bug log

Bug ID	Description	Severity	Status	Solution / Fix Implemented
BUG-01	Missing default Leaflet marker assets on production build	Medium	**Resolved**	Configured unpkg CDN marker icon overrides in `MapView.jsx`.
BUG-02	Cloud Firestore database returning 404 when unprovisioned	High	**Resolved**	Implemented dual-mode storage layer; app automatically runs in reactive local mode.
BUG-03	Bed counter decrements below zero on rapid clicking	Low	**Resolved**	Added `Math.max(0, current - 1)` boundary guard in `HospitalDashboard.jsx`.
BUG-04	Notification drawer backdrop not closing on mobile touch	Low	**Resolved**	Added touch backdrop listener in `NotificationsDrawer.jsx`.

---

8. Application Deployment

8.1 Cloud Deployment (Netlify Edge & GitHub)

The application is deployed live on Netlify's high-speed global CDN edge network:

* **Production URL:** `https://lifelinkviva.netlify.app/`

* **GitHub Repository:** `https://github.com/kaju2818/lifeLink`

* **Continuous Integration:** Netlify watches the `main` branch of `kaju2818/lifeLink` and executes automated production builds upon every Git commit.

8.2 Local Hosting

To execute LifeLink locally on any development PC:

```
# 1. Clone the repository
git clone https://github.com/kaju2818/lifeLink.git
cd lifeLink

# 2. Install dependencies
npm install

# 3. Launch Vite local development server
npm run dev
# Server ready at http://localhost:5173/
```

---

9. Performance & Security Testing

9.1 Basic Load Testing & Benchmarking

Table 32: Client-side benchmark

Metric / Measurement	Result Observed	Evaluation
DOM Content Loaded (Vite Dev / Local)	42 milliseconds	Extremely fast
Client-Side Route Transition (React Router)	< 15 milliseconds	Instantaneous
500 Donor Array In-Memory Filter Execution	2.1 milliseconds	Zero bottleneck
Production Bundle Size (HTML + JS + CSS)	~1.1 MB (includes complete Leaflet & Lucide icon sets)	Highly optimized

10. Final Documentation

10.1 Evaluation Against Objectives

Table 36: Evaluation against objectives

Objective	Evidence in Project	Status
Voluntary blood donor matching by group	Implemented in `/blood-finder` with 8 blood group filters.	**Achieved**
Real-time hospital bed & ICU vacancy tracking	Implemented in `/hospital-finder` and `/hospital/dashboard`.	**Achieved**
24/7 pharmacy medicine stock discovery	Implemented in `/medicine-finder` with ₹ pricing and prescription flags.	**Achieved**
1-Tap emergency SOS broadcast	Implemented in `/emergency` with GPS geocoding and 108 helpline dials.	**Achieved**
Role-based stakeholder access control	5 distinct portals guarded by `ProtectedRoute.jsx`.	**Achieved**

10.2 User Manual Summary

1. **Public Search:** Open `https://lifelinkviva.netlify.app/`. Click any discovery tab (Blood, Beds, Medicines).

2. **Emergency SOS:** Click the red **"EMERGENCY SOS"** button in the header or call **108** directly.

3. **Login & Role Portals:** Go to `/login` and select any 1-click demo account (Recipient, Donor, Hospital, Pharmacy, Admin) to immediately evaluate role dashboards.

10.3 Conclusion & Future Scope

LifeLink successfully establishes a high-speed, zero-cost, real-time healthcare availability platform that mitigates fatal delays during acute medical crises.

Future Roadmap:

1. Direct integration with IoT smart hospital bed pressure sensors.

2. Automated SMS / WhatsApp emergency alerts via Twilio / Gupshup gateways.

3. Interoperability with India's Ayushman Bharat Digital Mission (ABDM / ABHA ID).

4. Machine learning models for predictive regional blood shortage forecasting.

APPENDIX E: DATA DICTIONARY AND GLOSSARY

E.1 Data Dictionary

Table 41: Data dictionary

Entity / Field	Data Type	Description
`users.id`	string (UUID)	Unique user identification key.
`users.role`	string	Access role: `USER`, `DONOR`, `HOSPITAL`, `PHARMACY`, `ADMIN`.
`donors.bloodGroup`	string	Blood phenotype: `A+`, `A-`, `B+`, `B-`, `AB+`, `AB-`, `O+`, `O-`.
`donors.availability`	string	`AVAILABLE` (ready for dispatch) or `RECENTLY_DONATED` (resting).
`hospitals.beds.icu`	object `{total, available}`	ICU capacity and current vacancy count.
`medicines.price`	float	Unit Maximum Retail Price in Indian Rupees (₹).
`emergency.status`	string	`ACTIVE` (in transit) or `ACCEPTED` (hospital confirmed).

E.2 Glossary

Table 43: Glossary

Term	Meaning
Golden Hour	The first hour following traumatic injury where immediate medical intervention drastically reduces mortality.
ICU	Intensive Care Unit: specialized hospital facility for critically ill patients requiring continuous monitoring.
SDP	Single Donor Platelets: blood component collected via apheresis for severe dengue or oncology emergencies.
RBAC	Role-Based Access Control: restricting system features based on verified user identities.
GIS	Geographic Information System: spatial framework for mapping and geographical analysis.
SPA	Single Page Application: web application that rewrites the current web page with new data from the server.
Nominatim	Open-source search and reverse geocoding engine for OpenStreetMap data.
BaaS	Backend-as-a-Service: cloud architecture providing managed authentication and database persistence.
MoSCoW	Must, Should, Could, Won't: prioritization technique used in software requirement engineering.